Test Plan & QA Document

ChequePrint Pro — Quality Assurance

Version: 1.0.0
Testing Scope: Unit Tests, Integration Tests, End-to-End Tests, UAT

1. Test Scope

Area	Included
Number-to-words conversion	✅ Yes
Zod validation schemas	✅ Yes
SQLite database operations	✅ Yes
CSV/Excel import parsing	✅ Yes
Print HTML generation	✅ Yes
Electron IPC handlers	✅ Yes
UI component rendering	✅ Yes
End-to-end print flow	✅ Yes
Printer hardware integration	⚠ Manual only
Multi-OS compatibility	⚠ Manual only

2. Unit Tests

2.1 numberToWords Module

```
// __tests__/numberToWords.test.js
describe('numberToIndianWords', () => {
  test('zero', () => expect(numberToIndianWords(0)).toBe('Zero Only'));
  test('single digit', () => expect(numberToIndianWords(5)).toBe('Five Only'));
  test('teens', () => expect(numberToIndianWords(15)).toBe('Fifteen Only'));
  test('tens', () => expect(numberToIndianWords(70)).toBe('Seventy Only'));
  test('hundreds', () => expect(numberToIndianWords(500)).toBe('Five Hundred Only'))
  test('thousands', () => expect(numberToIndianWords(5000)).toBe('Five Thousand Only'
  test('thousands with remainder', () =>
    expect(numberToIndianWords(75050)).toBe('Seventy Five Thousand And Fifty Only'));
  test('lakhs', () =>
    expect(numberToIndianWords(100000)).toBe('One Lakh Only'));
  test('lakhs with thousands', () =>
    expect(numberToIndianWords(125000)).toBe('One Lakh Twenty Five Thousand Only'));
  test('crores', () =>
```

```
    expect(numberToIndianWords(10000000)).toBe('One Crore Only'));
  test('crores with lakhs and thousands', () =>
    expect(numberToIndianWords(15250000)).toBe('One Crore Fifty Two Lakh Fifty Thousa
  test('max amount', () =>
    expect(numberToIndianWords(999999999)).toBe(
      'Ninety Nine Crore Ninety Nine Lakh Ninety Nine Thousand Nine Hundred And Nine
    ));
});
```

2.2 Validation Schema

```
// __tests__/validation.test.js
describe('chequeSchema', () => {
  const validData = {
    payee_name: 'Rajesh Kumar',
    amount: 25000,
    cheque_date: '15/04/2026',
    bank_code: 'HDFC'
  };

  test('valid cheque passes', () => {
    expect(() => validateCheque(validData)).not.toThrow();
  });

  test('empty payee fails', () => {
    expect(() => validateCheque({ ...validData, payee_name: '' })).toThrow();
  });

  test('zero amount fails', () => {
    expect(() => validateCheque({ ...validData, amount: 0 })).toThrow();
  });

  test('amount over limit fails', () => {
    expect(() => validateCheque({ ...validData, amount: 1000000000 })).toThrow();
  });

  test('wrong date format fails', () => {
    expect(() => validateCheque({ ...validData, cheque_date: '2026-04-15' })).toThrow
  });

  test('stale date (>3 months ago) fails', () => {
    expect(() => validateCheque({ ...validData, cheque_date: '01/01/2025' })).toThrow
  });

  test('future date passes', () => {
    expect(() => validateCheque({ ...validData, cheque_date: '15/06/2026' })).not.toT
  });

  test('valid IFSC passes', () => {
    const data = { ...validData, ifsc_code: 'HDFC0001234' };
    expect(() => validateCheque(data)).not.toThrow();
  });

  test('invalid IFSC fails', () => {
    const data = { ...validData, ifsc_code: 'INVALID123' };
```

```
    expect(() => validateCheque(data)).toThrow();
  });

  test('payee name over 80 chars fails', () => {
    const data = { ...validData, payee_name: 'A'.repeat(81) };
    expect(() => validateCheque(data)).toThrow();
  });
});
```

2.3 CSV Import

```
// __tests__/csvImport.test.js
describe('validateRows', () => {
  test('valid row passes', () => {
    const rows = [{ payee_name: 'Test', amount: '5000', date: '15/04/2026' }];
    const result = validateRows(rows);
    expect(result[0].valid).toBe(true);
  });

  test('missing payee_name flagged', () => {
    const rows = [{ amount: '5000', date: '15/04/2026' }];
    const result = validateRows(rows);
    expect(result[0].valid).toBe(false);
    expect(result[0].errors).toContain('payee_name is required');
  });

  test('non-numeric amount flagged', () => {
    const rows = [{ payee_name: 'Test', amount: 'abc', date: '15/04/2026' }];
    const result = validateRows(rows);
    expect(result[0].valid).toBe(false);
  });

  test('wrong date format flagged', () => {
    const rows = [{ payee_name: 'Test', amount: '5000', date: '2026-04-15' }];
    const result = validateRows(rows);
    expect(result[0].valid).toBe(false);
  });
});
```

3. Integration Tests

3.1 Database Integration

```
// __tests__/db.integration.test.js
describe('SQLite Operations', () => {
  let db;

  beforeAll(() => {
    db = initTestDatabase(':memory:'); // In-memory SQLite for tests
  });
```

```
  test('saves cheque record successfully', async () => {
    const result = await saveRecord(db, {
      bank_code: 'SBI',
      payee_name: 'Test Payee',
      amount: 10000,
      amount_words: 'Ten Thousand Only',
      cheque_date: '15/04/2026',
      status: 'printed'
    });
    expect(result.success).toBe(true);
    expect(result.id).toBeGreaterThan(0);
  });

  test('queries records by payee name', async () => {
    const result = await queryRecords(db, { payee: 'Test Payee' });
    expect(result.records.length).toBeGreaterThan(0);
    expect(result.records[0].payee_name).toBe('Test Payee');
  });

  test('cancels a record', async () => {
    const { id } = await saveRecord(db, { /* ... */ });
    await cancelRecord(db, id, 'Wrong amount entered');
    const { records } = await queryRecords(db, {});
    const record = records.find(r => r.id === id);
    expect(record.status).toBe('cancelled');
    expect(record.cancel_reason).toBe('Wrong amount entered');
  });
});
```

4. End-to-End Test Checklist

4.1 Manual E2E Test Cases

Test ID	Scenario	Steps	Expected Result	Pass/Fail
E2E-001	Print single SBI cheque	1. Select SBI, 2. Fill form, 3. Click Print	Cheque prints; record saved	—
E2E-002	Print single HDFC cheque	Same as above with HDFC	Cheque prints in correct positions	—
E2E-003	Batch print 5 cheques from CSV	Import CSV with 5 valid rows, print all	All 5 cheques print; 5 records saved	—
E2E-004	Calibration wizard	Complete all 4 steps	Offset saved; subsequent print uses offset	—
E2E-005	Cancel a cheque record	Find record, cancel with reason	Status changes to Cancelled	—
E2E-006	Export records to CSV	Filter by date, export	CSV file created with correct data	—
E2E-007	Custom template creation	Add new bank, design template, print	Print uses custom template positions	—

Test ID	Scenario	Steps	Expected Result	Pass/Fail
E2E-008	Import with errors	Import CSV with 2 valid rows + 1 invalid	Invalid row flagged; 2 valid rows printable	—
E2E-009	Stale cheque warning	Enter date 4 months ago	Warning shown; print blocked or confirmed	—
E2E-010	Settings persistence	Change default bank, restart app	Default bank still selected after restart	—

5. Printer Hardware Test Matrix

Test on these printer models with Canon PIXMA MG2500 as primary target:

Printer	OS	Status	Notes
Canon PIXMA MG2500	Windows 10	☒ Primary	Full test required
Canon PIXMA MG2500	Windows 11	☑ Required	
HP DeskJet 2700	Windows 10	☑ Required	
Epson L3250	Windows 10	☑ Required	
Canon PIXMA MG2500	macOS 12	⚠ Secondary	
Generic PCL Laser	Windows 10	⚠ Secondary	

For each printer, verify:

- [ ] Custom paper size (cheque dimensions) accepted
- [ ] Calibration offset applied correctly
- [ ] Text not clipped at edges
- [ ] Print quality sufficient at 300 DPI

6. Performance Test Cases

Test	Method	Target	Acceptable
App cold start	Timer from launch to UI ready	< 2 sec	< 3 sec
Records query (10,000 rows)	Stopwatch	< 200ms	< 500ms
CSV import (500 rows)	Stopwatch	< 1 sec	< 3 sec
Print job dispatch	Timer to print start	< 1.5 sec	< 2 sec
Preview render on field change	Frame timer	< 30ms	< 50ms
Installer size	File size check	< 90 MB	< 120 MB

7. Security Test Cases

Test	Expected Result
No outgoing network connections	Verified via network monitor
SQLite file not readable without app	File not plain text (WAL journaling)
Backup file AES encryption	Backup cannot be opened without password
No logging of sensitive data (amounts, payees) to console	Verified
Input sanitization (XSS in payee name)	HTML escaped in print document

8. UAT (User Acceptance Testing) Checklist

Test with 5 target users (2 small business owners, 2 accountants, 1 individual):

- [ ] User can install without IT support
- [ ] User can print first cheque in < 5 minutes (no calibration)
- [ ] Calibration wizard understood and completed independently
- [ ] Batch import from their existing Excel file works
- [ ] Records search returns expected results
- [ ] User rates overall experience ≥ 4/5